BIBEK SUBEDI

Prospective Graduate Student • Hydrology and Flood Hazard • Earthquake Engineering • Machine Learning

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Research Interests

Flood and debris-flow hazard in mountain river systems, hydrological modelling and model diagnostics with machine learning, design rainfall and extreme-value statistics under nonstationarity, and earthquake site response and slope hazard, using numerical models, satellite remote sensing and Python.

Education

Bachelor of Engineering in Civil Engineering 2021 – 2026

IOE Pulchowk Campus, Tribhuvan University, Lalitpur, Nepal

Higher Secondary Education in Science 2019 – 2021

SOS Hermann Gmeiner Secondary School, Bharatpur, Nepal *GPA:* ***3.78 / 4.00***

Publications

Journal articles, submitted

1. **Subedi, B.**, Karki, K., & Sijapati, M. K. Residual learning as a model diagnostic: discharge correction and recession mismatch in a calibrated SWAT model, Arun basin, eastern Nepal. Submitted to *Hydrological Sciences Journal*, 2026.
2. **Subedi, B.**, Bista, M., & Suwar, A. Storm-drain design rainfall in Indian million-plus cities: gauge-network inhomogeneity and the reliability of decadal updates. Submitted to *Journal of Hydrologic Engineering* (ASCE), 2026.

Conference papers, accepted October 2026: GeoMandu 2027 (11th AYGEC and 1st SACG), Springer proceedings

3. **Subedi, B.** A borehole-driven nonlinear Newmark atlas of earthquake-induced slope displacement in the Kathmandu Valley.
4. **Subedi, B.**, & Acharya, J. Saltelli–Sobol global sensitivity of nonlinear site amplification under the 2015 Gorkha earthquake motion.
5. **Subedi, B.**, Gautam, H., Jaishi, H., & Shah, K. R. Optimal siting of earthquake early-warning sensors in the Kathmandu Valley using multi-criteria decision analysis.
6. Karki, K., Sijapati, M. K., & **Subedi, B.** Nonlinear site-response assessment of rubber–soil caps and cushions for geotechnical seismic isolation.

Conference abstract, submitted

7. **Subedi, B.**, Bista, M., & Suwar, A. Changes in the 5-year design rainfall for urban storm-water drains in India's 52 million-plus cities, 1975–2024. Extended abstract submitted to INFRASURE 2027, IIT Kharagpur.

Research Experience

Rasuwa Rock-Ice Avalanche and Debris Flood: Reconstruction and Exposure Aug 2026 – Present

Lende–Bhote Koshi–Trishuli–Narayani river system, Nepal

- Reconstructed the 26 August 2026 debris flood triggered by a rock-ice avalanche in the Lende valley with HEC-RAS 6.6 two-dimensional shallow-water models on conditioned 12.5 m terrain, from the Nepal–China border at Rasuwagadhi to Devghat (195 km), with Python automation, r.avaflow mass-balance checks and HAND-based inundation screening.
- Evaluated runs against a satellite-mapped flood footprint (best critical success index about 0.72) and six DHM gauge timings, using Sentinel-1 SAR (Google Earth Engine), Sentinel-2, Landsat and differenced 2 m pre/post-event DEMs; constrained the release volume from the mapped footprint with debris-flow scaling relations.
- Screened exposure of buildings, roads and hydropower assets by ward and district and tested debris-dam breach scenarios.

Residual Learning as a Hydrological Model Diagnostic, Arun Basin 2026

Lead author | Transboundary Himalayan catchment, eastern Nepal

- Calibrated a multi-site SWAT model (SWAT-CUP, SUFI-2) of the 33,400 km^2 Arun basin and corrected its daily discharge with gradient boosting under nested leave-one-year-out validation, with LSTM and ridge baselines, block-bootstrap tests and SHAP attribution.
- Raised Nash–Sutcliffe efficiency from 0.53–0.66 to 0.70–0.81 at three nested gauges (63 of 70 station-years improved); showed that small pooled biases hid 21–33% overestimation of the lowest flows, and that observed recessions were 4.0–7.6 times faster than simulated.
- Configured and calibrated mHM 5.13/mRM for the basin (DDS, 37 parameters; KGE 0.847 in calibration, 0.836 in validation) after testing ERA5-Land, APHRODITE, GLDAS, TerraClimate and IMERG forcing against the catchment

water balance and MODIS snow cover, in correspondence with Dr Pallav Kumar Shrestha (UFZ Leipzig).

Design Rainfall for Urban Storm Drains, Indian Million-Plus Cities 2026

Lead author | IMD 0.25° gridded daily rainfall, 1951–2024

- Located a gauge-network jump in 1975 with break tests, fitted GEV distributions by L-moments with bootstrap intervals for the 5-year design depth of 52 cities, and separated the step from trend with a nonstationary model; Monte Carlo simulation showed that the national manual's decadal-update rule raises design depths in half of trend-free records, and a one-sided likelihood-ratio test kept false alarms near 7%.

Earthquake-Induced Slope Displacement and Seismic Hazard, Kathmandu Valley May 2025 – Present

Research Cohort Member, Kathmandu Geo Lab

- Developed a borehole-driven nonlinear Newmark atlas of earthquake-induced slope displacement for the valley; first author of a Saltelli–Sobol sensitivity analysis of nonlinear site amplification (6,144 OpenSees runs) and of multi-criteria (GIS-AHP, maximal-coverage) siting of early-warning sensors; all accepted for GeoMandu 2027.
- Contributed to shear-wave velocity prediction with random forest, gradient boosting and support vector regression.

Seismic Microzonation of the Kathmandu Valley 2025 – 2026

IOE Pulchowk Campus | Non-linear 1D site response using OpenSees

- Compiled 332 borehole profiles and ran non-linear 1D site-response analyses (PDMY/PIMY) under six input ground motions, including the 2015 Gorkha mainshock (1,992 analyses); produced GIS-ready amplification maps and compared them with code site factors.

Skills and Languages

Flood and hazard modelling: HEC-RAS 6.6 (1D/2D), r.avaflow, HAND inundation screening, exposure mapping, debris-dam breach scenarios, Newmark slope displacement.

Hydrological modelling: SWAT/SWAT-CUP, mHM/mRM; calibration and validation, catchment water balance, flood-frequency and extreme-value analysis (GEV, L-moments, nonstationary GEV).

Machine learning and statistics: gradient boosting (LightGBM), LSTM, random forest, SVR, ridge regression, SHAP; block bootstrap, break and likelihood-ratio tests, Sobol and Monte Carlo analysis.

Remote sensing and GIS: Google Earth Engine; Sentinel-1 SAR (flood extent, SNAP DInSAR), Sentinel-2, Landsat, MODIS; DEM differencing, terrain analysis; QGIS, ArcGIS Pro, GRASS GIS, GDAL/OGR.

Geotechnical and earthquake engineering: OpenSees, PLAXIS; 1D/2D site response, nonlinear effective-stress analysis, seismic site classification.

Computing: Python (NumPy, pandas, SciPy, GeoPandas, rasterio), C, Bash, Linux/WSL, Git, LaTeX, MS Excel.

Languages: Nepali (native), English and Hindi.

Field, Professional and Community Work

- **Intern, Geospatial and Field Engineering, Kathmandu Upatyaka Khanepani Limited (KUKL)** (2026), internship theme *Integration of Water Supply Connection Details and Numbering of Buildings inside Singha Durbar using Geographic Information Systems*: field surveys to verify water-supply connections and building numbers against records, and a GIS database for utility planning.
- **Project Lead, Eco-Friendly Sanitation, WorldLink CSR, Bhadaure, Dhading** (Jan 2025 – Present): co-ran a baseline survey of all 25 households on remote, sloping terrain and prepared an NPR 1.57 million Phase-I design and cost estimate with rooftop rainwater harvesting; also **Design and Costing Lead**, community park plantation with the Manjushree CFUG, Chobhar (2026).

Leadership, Teaching and Outreach

- **Research Wing Correspondent, CESS-Nepal** (Jul 2025 – Present): organised a 15-day machine learning and AI workshop for civil engineering students; Event Coordinator, Civil Engineering Hackathon (2024 – 2025). **Vice President, MAPS Pulchowk** (Dec 2023 – Dec 2025): directed the Integration Bee (500+ participants) and Physics and Mathematics Olympiads.
- **IOE Entrance Mentor, Clamphook Academy** (2021 – Present): 25,000+ students mentored, including scholarship students from low-income families; **Educational Creator, YouTube** (3,000+ subscribers).

Presentations, Training and Recognition

- Three posters at the Kathmandu Geo Lab Student Research Symposium with Duke University Bass Connections, Lalitpur, 22 May 2026: nonlinear site response and co-seismic slope displacement (first author), stochastic site response and earthquake early-warning sensor placement.
- Winner-Mentor, 3rd GeoMandu Geotechnical Model Competition, Nepal Geotechnical Society (2026); Flood Hazard and Exposure Mapping, UNESCO Open Learning (2025); Machine Learning and AI with Python, 15-day workshop, CESS-Nepal (2025 – 2026); Software Fellowship, LOCUS 2026, IOE Pulchowk.